Star Schema Diagram Documentation

AeroGrid

Low-Altitude Airspace Intelligence Warehouse

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1 Schema Architecture

The AeroGrid Data Warehouse utilizes a Star Schema architecture optimized for OLAP aggregations. The core structure centers around five fact tables radiating out to shared dimension tables.

2 Core Dimensions and Star Topology

- **Star Core (`DimTime`):** Connected to all five fact tables via `time_key`, enabling uniform temporal filtering and drill-down analysis.
- **Actor Star (`DimVehicle`):** Connected to tracking, journey, and close encounter facts. Uses role-playing foreign keys (`vehicle_key_1` and `vehicle_key_2`) for multi-vehicle conflict analysis in `FactCloseEncounter`.
- **Spatial Star (`DimSpatialGrid` and `DimAirspaceZone`):** Links position reports, close encounters, and occupancy snapshots to geographic grid cells and FAA restriction boundaries.

3 Schema Summary

The complete database model is formalized in DBML format (`data/diagrams/aerogrid_schema.dbml`) ensuring seamless visualization in tools like dbdiagram.io and automated schema migration.